Type MS · Name receptors and the senses or stimuli they detect

Answer 1 of 4 · Question 1 part c.iii · 1 mark

2024 JUN · Paper 01 · find part c.iii in this mark scheme extract

1 (a)	(i)	sensory (neurone/cell);		(1)
	(ii)	axon;		(1)
(b)	(i)	Part R sweat gland; Part S blood vessels/capillaries/arterioles;		(2)
	(ii)	Any two from: • produces sweat/perspiration; • sweat evaporates (from skins surface); • using heat energy (from the blood);		(2)
	(iii)	• provides an insulating layer insulation; • traps heat (energy)/to prevent heat loss;	reject keeps body warm	(2)
(c)	(i)	• water is too hot/burn hands;		(1)
	(ii)	• use a thermometer; • to test the temperature of the water/to make sure the water is not too hot;		(2)
	(iii)	• (temperature) receptor;		(1)

Total for Question 1 = 12 Marks

Type MS · Name receptors and the senses or stimuli they detect

Answer 2 of 4 · Question 4 part a · 2 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

Number				
4	(a)	any two from • hearing/sound; • touch; • taste; • smell;		2
	(b) (i)	any two from • as thickness decreases, focal length increases; • interval between focal lengths increases as thickness decreases; • (constant when two variables multiplied hence) inversely proportional;		2
	(ii)	diameter decreases;		1
	(iii)	any four from • rays from near object diverging; • ciliary muscle contracts; • suspensory ligaments slacken/relax; • so lens becomes thicker/more convex; • more powerful lens/light refracted more/increased refraction; • (allows it to focus because) focal length becomes shorter/decrease;		4
	(c) (i)	between 575 and 580 nm;		1
	(ii)	• 400 - 530; • 130 nm;	ECF	2
	(iii)	• greater range of wavelengths can be seen/absorb light of different wavelengths; • gives better colour perception/vision/differentiation/can see different colours;		2

Total 14

Type MS · Name receptors and the senses or stimuli they detect

Answer 2 of 4 · Question 4 part a · 2 marks

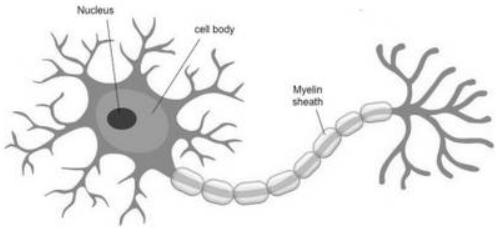
2023 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Name receptors and the senses or stimuli they detect

Answer 3 of 4 · Question 7 part a.i · 1 mark

2024 JUN · Paper 01 (regional R) · find part a.i in this mark scheme extract

7 (a) (i)	receptor (cell)		(1)
(ii)	sensory neurone		(1)
(b) (i)	a cell that has particular features/adaptations/structures; to carry out a certain/specific function in the body;		(2)
(ii)	diagram; to include cell body at one end, with dendrites and an elongated axon twice the length of the cell body and axon ending; nucleus; cell body; myelin sheath;  	Label marks can be awarded if they draw a sensory or relay neurone	(4)
(iii)	transmits electrical impulses from the brain/CNS/spinal cord/ relay neurone; to effecter organs/muscle/gland;		(2)
(c)	100/4.0 x 10^{6}; = 0.000025 (s); = 0.025 (ms); = 2.5 x 10^{-2};	ecf for mps 2,3 and 4 Allow full marks for correct final answer	(4)

Total for Question 7 = 14 marks

Type MS · Name receptors and the senses or stimuli they detect

Answer 4 of 4 · Question 4 part c.iii · 2 marks

2023 JUN · Paper 01 · extract 1 of 2 · find part c.iii in this mark scheme extract

Number				
4	(a)	any two from • hearing/sound; • touch; • taste; • smell;		2
	(b) (i)	any two from • as thickness decreases, focal length increases; • interval between focal lengths increases as thickness decreases; • (constant when two variables multiplied hence) inversely proportional;		2
	(ii)	diameter decreases;		1
	(iii)	any four from • rays from near object diverging; • ciliary muscle contracts; • suspensory ligaments slacken/relax; • so lens becomes thicker/more convex; • more powerful lens/light refracted more/increased refraction; • (allows it to focus because) focal length becomes shorter/decrease;		4
	(c) (i)	between 575 and 580 nm;		1
	(ii)	• 400 - 530; • 130 nm;	ECF	2
	(iii)	• greater range of wavelengths can be seen/absorb light of different wavelengths; • gives better colour perception/vision/differentiation/can see different colours;		2

Total 14

Type MS · Name receptors and the senses or stimuli they detect

Answer 4 of 4 · Question 4 part c.iii · 2 marks

2023 JUN · Paper 01 · extract 2 of 2 · find part c.iii in this mark scheme extract

Question number	Answer	Notes	Marks